

Title:

Task-Conditioned Object Detection for FPGA-
Based Edge Deployment

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Approach Description

This project implements a lightweight task-aware object selection system on the PYNQ-Z2 using YOLOv4-tiny and the COCO dataset. Unlike traditional object detection systems that simply identify all objects in a scene, our approach focuses on selecting the most suitable object for a specific task. The system first displays a list of predefined tasks such as “serve wine”, “open parcel”, and “water plant”. After the user selects a task, the webcam captures an image and the YOLOv4-tiny model detects objects present in the scene. The detection model outputs multiple object candidates along with their confidence scores and bounding boxes. Instead of displaying all detections as the final output, the proposed system performs task-aware affordance reasoning. For every task, a semantic mapping table was manually defined, assigning priority values to relevant objects. These priorities represent the suitability of an object for performing a specific task. For example, for the task “serve wine”, a wine glass is assigned the highest priority, followed by a cup and then a bottle. Similarly, for “open parcel”, scissors are given higher importance than a knife.

The final selection score is computed by multiplying the object detection confidence score with the predefined affordance priority value. This combined score enables the system to select the object that is both visually confident and semantically appropriate for the chosen task. Among all detected task-relevant objects, only the object with the highest final score is highlighted as the best object for that task. This transforms the system from generic object detection into a goal-directed visual reasoning system.

The original research paper “What Object Should I Use? – Task Driven Object Detection” proposed a deep learning-based task-driven object detector trained using specialized datasets and GPU-based architectures such as Faster R-CNN. However, the exact approach from the paper could not be implemented directly on the PYNQ-Z2 platform due to hardware and software limitations. The original implementation depends heavily on PyTorch, CUDA acceleration, large GPU memory, and computationally expensive training pipelines. In contrast, the PYNQ-Z2 board contains a dual-core ARM Cortex-A9 processor with limited RAM and no GPU acceleration. Additionally, official PyTorch support for the ARMv7 architecture and Python version available on the board is not practically feasible.

Because of these limitations, a lightweight approximation of the paper’s approach was developed instead. The proposed implementation preserves the core concept of task-driven object selection while replacing computationally expensive deep learning reasoning modules with manually designed affordance priors and efficient YOLOv4-tiny inference through OpenCV DNN. This adaptation enabled real-time embedded execution while maintaining the semantic behavior of task-conditioned object selection. Consequently, the developed system serves as an embedded-friendly approximation of task-driven object detection suitable for FPGA-based and low-power edge AI platforms.

Design Contest Stage 2(A) Report

Results

Not Secure 192.168.2.2:9090/notebooks/DVCon.ipynb

Jupyter DVCon Last Checkpoint: 3 hours ago (unsaved changes)

File Edit View Insert Cell Kernel Widgets Help Trusted Python 3 (ipykernel)

Select Task Number: 10

Selected Task: serve wine
Camera opened successfully

[WARN:0] global ./modules/videoio/src/cap_gstreamer.cpp (616) isPipelinePlaying OpenCV | GStreamer warning: GStreamer: pipeline have not been created

Image captured successfully
cup | confidence=0.96 | priority=0.90 | final=0.86

BEST OBJECT: cup



Not Secure 192.168.2.2:9090/notebooks/DVCon.ipynb

Jupyter DVCon Last Checkpoint: 3 hours ago (unsaved changes)

File Edit View Insert Cell Kernel Widgets Help Trusted Python 3 (ipykernel)

Select Task Number: 11

Selected Task: pour sugar
Camera opened successfully

[WARN:0] global ./modules/videoio/src/cap_gstreamer.cpp (616) isPipelinePlaying OpenCV | GStreamer warning: GStreamer: pipeline have not been created

Image captured successfully
cup | confidence=0.94 | priority=0.30 | final=0.28
spoon | confidence=0.54 | priority=1.00 | final=0.54

BEST OBJECT: spoon

